EXCISION OF MODIFICATIONS AT A HENSELIAN LOCAL SCHEME POINT

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4480-4512, following the statement at 4455-4478.

Editorial note. No omissions; the henselization-as-colimit and its Spec are anchored to the same cofiltered index category.

Allowed standard prerequisites (editorial).

- Elementary etale neighbourhoods of decent spaces. Decent Spaces, Lemma decent-spaces-lemma-decent-space-elementary-etale-neighbourhood (cross-chapter, cited at ~4481-4484). Affine elementary etale neighbourhood (W, w) of the closed point x.
- Excision equivalence for modifications. lemma-excision-modifications and lemma-excision-modifications-properties (same chapter, card L26302, cited at ~4490-4491). Reduces to the affine scheme case.
- Cofilteredness of elementary etale neighbourhoods. Decent Spaces, Lemma decent-spaces-lemma-elementary-etale-neighbourhoods or More on Morphisms, Lemma more-morphisms-lemma-elementary-etale-neighbourhoods (cross-chapter, cited at ~4499-4502). The index category of neighbourhoods is cofiltered.
- Directed limits with affine transitions exist. lemma-directed-inverse-system-has-limit (same chapter, card L26291, cited at ~4503-4505). Spec(O^h_{X,x}) = lim U and V = lim U setminus {u}.
- Descent of finite presentation morphisms. lemma-descend-finite-presentation (same chapter, card L26286, cited at ~4507-4508). The right category is the colimit of the categories for (U, u).

Source reference. Original reference 0BBP: focused-seeds/geometry/S9-spaces-limits-modifications/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0BGV: focused-seeds/geometry/S9-spaces-limits-modifications/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SK: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2148–2148. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BGY: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 4337–4337. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0BGZ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 4420–4420. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 057B: focused-seeds/geometry/S9-spaces-limits-modifications/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Let $x \in |X|$ be a closed point such that $U = X \setminus \{x\} \to X$ is quasi-compact. With $V = \mathrm{Spec}(\mathcal{O}^h_{X,x}) \setminus \{\mathfrak{m}^h_x\}$ the base change functor*

$$\left\{\begin{array}{l} f : Y \to X \text{ of finite presentation} \\ f^{-1}(U) \to U \text{ is an isomorphism} \end{array}\right\} \longrightarrow \left\{\begin{array}{c} g : Y \to \mathrm{Spec}(\mathcal{O}^h_{X,x}) \text{ of finite presentation} \\ g^{-1}(V) \to V \text{ is an isomorphism} \end{array}\right\}$$

is an equivalence of categories.

Proof. Let $a : (W, w) \to (X, x)$ be an elementary étale neighbourhood of x with W affine as in Decent Spaces, Lemma 0BBP. Since x is a closed point of X and w is the unique point of W lying over x, we see that w is a closed point of W. Since a is étale and identifies residue fields at x and w, it follows that a induces an isomorphism $a^{-1}x \to x$ (as closed subspaces of X and W). Thus we may apply Lemma 0BGY and 0BGZ to reduce the problem to the case where X is an affine scheme.

Assume X is an affine scheme. Recall that $\mathcal{O}^h_{X,x}$ is the colimit of $\Gamma(U, \mathcal{O}_U)$ over affine elementary étale neighbourhoods $(U, u) \to (X, x)$. Recall that the category of these neighbourhoods is cofiltered, see Decent Spaces, Lemma 0BGV or More on Morphisms, Lemma 057B. Then $\mathrm{Spec}(\mathcal{O}^h_{X,x}) = \lim U$ and $V = \lim U \setminus \{u\}$ (Lemma 07SF) where the limits are taken over the same category. Thus by Lemma 07SK The category on the right is the colimit of the categories for the pairs (U, u). And by the material in the first paragraph, each of these categories is equivalent to the category for the pair (X, x). This finishes the proof. □